NUMBER & ALGEBRA

CHAPTER 3

Proofs

SYLLABUS 1.6 1.15

Mark some points on a circle. Join every pair of points with a straight line. Then count the regions inside the circle. Place the points carefully, so that no three lines cross at the same spot.

Count the regions for the first few cases.

Points n	**1**	**2**	**3**	**4**	**5**
Regions	1	2	4	8	16

The pattern seems clear: each new point doubles the number of regions. So six points must give 32.

Six points give **31**.

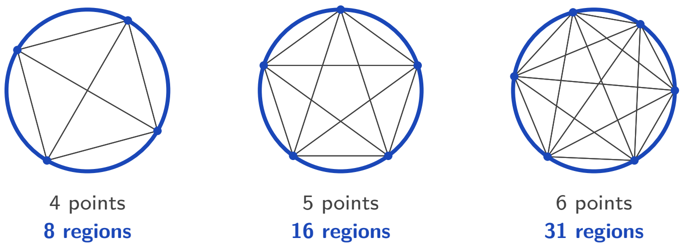

Figure 3.1 Four points give 8 regions and five give 16, but six give 31, not 32.

Five cases followed the doubling rule and the sixth did not. The first five gave no sign of it, and checking further cases would not have helped.

This is the difference between seeing a pattern and knowing that it always holds. A pattern is a guess. The word for such a guess is a **conjecture**: a statement believed true but not yet proved. The doubling rule was a conjecture, and the sixth case disproved it.

A **proof** is an argument in which each step follows from the one before, so the conclusion cannot fail, however many points you take.

By the end of this chapter you will be able to prove a statement directly from definitions, disprove one with a single counterexample, argue by contradiction, and prove a result about every positive integer by induction. The course asks for this because a formula is only trustworthy once it has been proved. Checking a rule on a few cases shows that it has not failed yet. A proof shows that it cannot fail.

3.1 Deductive Proof

A **proof** is a chain of logical steps. It starts from things already accepted, such as definitions, results proved earlier, and the rules of algebra, and it ends at the statement you want to establish. Each step must follow with certainty from the steps before it. If the chain is correct, the conclusion is not merely likely. It cannot be false.

This is **deductive** reasoning: you apply general principles to reach a specific, guaranteed result. The oldest example is not mathematical at all:

- Every human being is mortal. (a general principle)
- Socrates is a human being. (a specific case)
- Therefore Socrates is mortal. (the conclusion)

The conclusion is not a guess, and no amount of further evidence could improve it. If the first two statements are true, the third cannot be false. Mathematics uses arguments of exactly this shape:

- Every multiple of 4 is even.
- The number 132 is a multiple of 4.
- Therefore 132 is even.

Notice that you did not divide 132 by 2 to check. The conclusion was forced by the two statements above it.

The circle problem went the other way. It used a few particular cases to guess a general rule. Reasoning in that direction, from examples to a general claim, is called induction in the everyday sense of the word, and it can lead to a false conclusion. Deduction cannot.

Equality versus identity

Before you prove a statement about an equation, you must know which kind of statement it is. The equals sign is used for two very different claims.

DEF An **equation** is a statement that two expressions are equal for *some particular* values of the variable. An **identity** is a statement that they are equal for *every* value for which both sides are defined. An identity is written with the symbol $\equiv$:

$$(x+1)^2 \equiv x^2 + 2x + 1.$$

The equation $x^2 = 4$ is true only when $x = 2$ or $x = -2$. The identity $(x+1)^2 \equiv x^2 + 2x + 1$ is true for every real x. Solving an equation means finding the values that make it true. Proving an identity means showing that no value makes it false. The three-bar symbol $\equiv$ tells you which of the two you are dealing with: when you see it, the claim is "for all x", not "find x".

You already know one important identity from trigonometry:

$$\sin^2 x + \cos^2 x \equiv 1.$$

This is true for every angle x, not for some particular angles, which is why it is written with $\equiv$. It is proved and used in Ch. 9.

Proving an identity

The two sides of an identity have names. The expression to the left of the $\equiv$ sign is the **left-hand side**, written **LHS**. The expression to the right is the **right-hand side**, written **RHS**.

To prove an identity you take one side and transform it, step by step, into the other, using only valid algebra. Start from the more complicated side and simplify it: write down the LHS, change it one line at a time, and finish with the RHS. Write the labels LHS and RHS in your working, so the direction of the argument is clear to the marker.

CAUTION *Do not start by assuming the identity is true and then manipulate both sides until you reach something obvious like* $0 = 0$*. That assumes what you are trying to prove. Work one side into the other instead.*

EXAMPLE 3.1

The difficulty here is avoiding the common error of doing the same thing to both sides. Prove that $(a+b)^2 - (a-b)^2 \equiv 4ab$.

Start from the LHS and expand each square:

$$\text{LHS} = (a+b)^2 - (a-b)^2 = (a^2 + 2ab + b^2) - (a^2 - 2ab + b^2).$$

Now remove the brackets. Take care with the signs in the second bracket:

$$= a^2 + 2ab + b^2 - a^2 + 2ab - b^2 = 4ab = \text{RHS}. \quad ■$$

The LHS became the RHS using nothing but expansion. At no point did we assume the two sides were equal; we discovered it.
A quick **check** takes five seconds and finds most errors: at $a = 3$, $b = 2$ the left side is $25 - 1 = 24$ and the right side is $4(3)(2) = 24$. A check is not a proof. But if a check fails, you know for certain that something is wrong.

EXAMPLE 3.2
Show that $(x-3)^2 + 5 \equiv x^2 - 6x + 14$.
The LHS is the more complicated side, so start there and expand the square:

$$\text{LHS} = (x-3)^2 + 5 = (x^2 - 6x + 9) + 5.$$

Collect the constant terms:

$$= x^2 - 6x + 14 = \text{RHS}. \quad \blacksquare$$

Check at $x = 1$: the LHS is $(-2)^2 + 5 = 9$, and the RHS is $1 - 6 + 14 = 9$.

TIP *Always check your own results by substituting a number, in proofs and everywhere else. Examiners expect you to be able to verify a result you have just derived, and one well-chosen value will reveal an algebra error immediately.*

EXAMPLE 3.3
Show that $\frac{1}{4} + \frac{1}{12} = \frac{1}{3}$, and prove the algebraic generalisation

$$\frac{1}{m+1} + \frac{1}{m^2+m} \equiv \frac{1}{m}.$$

The numerical case is direct: $\frac{1}{4} + \frac{1}{12} = \frac{3}{12} + \frac{1}{12} = \frac{4}{12} = \frac{1}{3}$. Notice it is the claimed identity at $m = 3$: $m + 1 = 4$ and $m^2 + m = 12$.
For the general statement, start from the left side and factor the second denominator, $m^2+m = m(m+1)$:

$$\frac{1}{m+1} + \frac{1}{m(m+1)} = \frac{m}{m(m+1)} + \frac{1}{m(m+1)} = \frac{m+1}{m(m+1)} = \frac{1}{m}. \quad \blacksquare$$

One calculation confirmed a single case. The algebra proved every case at once, for every $m \neq 0, -1$. This "verify, then generalise" structure is exactly how such questions are phrased in examinations.

Proving a general statement

Most proofs are not about identities. They are about claims such as "the sum of two odd numbers is even". The first step is always the same: write the general object in algebra, so that one expression stands for every case at once.

The table below lists the standard forms. In every row, k and n are integers.

In words	In algebra
an even number	$2k$
an odd number	$2k+1$, or $2k-1$
two consecutive integers	n and $n+1$
three consecutive integers	$n-1$, n, $n+1$
two consecutive even numbers	$2k$ and $2k+2$
two consecutive odd numbers	$2k+1$ and $2k+3$
a multiple of 3	$3k$
a number that leaves remainder 1 when divided by 3	$3k+1$
a square number	k^2
a rational number	p/q, with p, q integers and $q \neq 0$

Choosing the right form matters. To prove a statement about *any* two odd numbers, use $2a+1$ and $2b+1$ with two different letters. Writing $2a+1$ twice would only prove the statement for two *equal* odd numbers.

Once the general case is written in symbols, ordinary algebra does the rest.

EXAMPLE 3.4

A single example such as $3+5=8$ confirms nothing about *all* odd numbers. Prove that the sum of any two odd numbers is even.

Let the two odd numbers be $2a+1$ and $2b+1$, where a and b are integers. Their sum is

$$(2a+1)+(2b+1) = 2a+2b+2 = 2(a+b+1).$$

Since $a+b+1$ is an integer, the sum is $2 \times (\text{integer})$, which is by definition even. ■

EXAMPLE 3.5

Prove that the product of two consecutive integers is always even.

Take the consecutive integers n and $n+1$. One of any two consecutive integers must be even: if n is even the product $n(n+1)$ has an even factor, and if n is odd then $n+1$ is even and again the product has an even factor. Either way $n(n+1)$ is a multiple of 2, so it is even. ■

CAUTION *One example can never prove a statement that begins "for all." It can only disprove one, as the next section shows. The proofs above work because the algebra stands for every case at once.*

YOUR TURN 3.1 Deductive Proof

1. State whether each of the following is an equation or an identity.

(a) $3x = 12$ [1]
(b) $(x+2)^2 = x^2 + 4x + 4$ [1]

2. Prove each identity.

(a) $(a+b)^2 = a^2 + 2ab + b^2$ [2]
(b) $(a-b)^2 = a^2 - 2ab + b^2$ [2]
(c) $\dfrac{1}{x} - \dfrac{1}{x+1} = \dfrac{1}{x(x+1)}$ [3]

3. Prove each statement about odd and even integers.

(a) The sum of any two even integers is even. [2]
(b) The sum of an even integer and an odd integer is odd. [2]
(c) The product of any two odd integers is odd. [2]

4. Prove each statement for every integer n.

(a) $(2n+1)^2$ is odd. [2]
(b) $n^2 - n$ is even. [2]

3.2 Disproof by Counterexample HL

A universal statement claims something about every member of a collection. To show that such a claim is false, you do not need to discuss the whole collection. You need one member for which the claim fails.

DEF A **counterexample** is one specific case for which a general statement fails. A single counterexample is enough to prove that the statement is false.

The logic is exact. "Every n has property P" is false as soon as one value of n does not have property P. This is what happened with the circle problem. The claim "the number of regions doubles each time" was a universal statement, and $n = 6$ was a counterexample to it.

EXAMPLE 3.6

A famous formula, $n^2 + n + 41$, produces a prime number for $n = 0, 1, 2, \ldots, 39$, forty primes in a row. Decide whether it produces a prime for *every* non-negative integer n.

Test the claim at the first value the pattern does not cover. At $n = 40$:

$$40^2 + 40 + 41 = 1600 + 40 + 41 = 1681 = 41^2.$$

This is 41×41, not prime. So $n = 40$ is a counterexample, and the statement "$n^2 + n + 41$ is always prime" is false. ∎

Forty cases supported the claim, and one case ended it. Examples can make a claim look true. They can never make it certain.

EXAMPLE 3.7

Disprove the claim: "every prime number is odd."

The number 2 is prime, and 2 is even. That single case is a counterexample, so the claim is false. ∎

EXAMPLE 3.8

Show that this statement is not always true: "there are no positive integer solutions to the equation $x^2 + y^2 = 10$."

Take $x = 1$ and $y = 3$. Both are positive integers, and

$$1^2 + 3^2 = 1 + 9 = 10.$$

So a positive integer solution does exist, and the statement is false. ∎

Notice what the answer contains: the values, the substitution, and the conclusion. Giving only "$x = 1$, $y = 3$" would not earn full marks.

TIP *A counterexample must be explained, not just named. State the case, substitute it into the claim, show the calculation, and say in words why this makes the claim false. The IB is explicit that stating the counterexample alone is not sufficient.*

Two more claims are worth testing before you read on. "If $a^2 > b^2$ then $a > b$" sounds obvious, but $a = -3$ and $b = 1$ give $9 > 1$ with $-3 < 1$, because squaring removes the sign. "The product of two irrational numbers is irrational" sounds reasonable, but $\sqrt{2} \times \sqrt{2} = 2$, since the two irrational parts cancel exactly.

RW In 1640 Fermat studied the numbers $F_n = 2^{2^n} + 1$ and found that $F_0 = 3$, $F_1 = 5$, $F_2 = 17$, $F_3 = 257$ and $F_4 = 65\,537$ are all prime. He conjectured that every F_n is prime. Almost a century later, in 1732, Euler tested the next one:

$$F_5 = 2^{32} + 1 = 4\,294\,967\,297 = 641 \times 6\,700\,417.$$

One counterexample ended the conjecture. No larger Fermat prime has ever been found.

CAUTION *A counterexample only disproves a claim. Finding no counterexample after many attempts is not a proof. The circle problem is the warning: five cases agreed, and the sixth did not.*

YOUR TURN 3.2 Disproof by Counterexample HL

5. Disprove each claim by giving a counterexample.

(a) Every integer is positive. [1]
(b) Every prime number is odd. [1]
(c) If n is divisible by 4 then n is divisible by 8. [1]
(d) The sum of two prime numbers is always even. [2]

6. Disprove each claim by giving a counterexample.

(a) $n^2 \geq 2n$ for all integers n. [2]
(b) $x^2 > x$ for all real x. [2]
(c) If n^2 is even then n is odd. [1]

3.3 Proof by Contradiction HL

Suppose a friend tells you they have written down the largest whole number that exists. You do not need to see it to know they are wrong. Whatever the number is, add 1 to it. The result is a whole number, and it is larger. The claim disproves itself.

You have just proved something without ever finding the answer. You did not search for the largest whole number. You assumed it existed and showed that the assumption leads to an impossible situation.

This is the idea of **proof by contradiction**. Assume the statement you want to prove is *false*. Reason correctly from that assumption until you reach something impossible. Correct reasoning cannot turn a true assumption into an impossible result, so the assumption must have been false. The original statement is therefore true.

DEF In **proof by contradiction**, you assume the negation of what you want to prove, then derive a logical contradiction. The contradiction forces the assumption to be false, which establishes the original claim.

The method has a Latin name, *reductio ad absurdum*, meaning "reduction to the absurd". It is powerful because it gives you something to work with. Proving that no fraction equals $\sqrt{2}$ directly would mean checking infinitely many fractions. Assuming one exists gives you an equation to manipulate. The two proofs below are among the most famous in mathematics, and both work this way.

EXAMPLE 3.9

Prove that $\sqrt{2}$ is irrational, that is, it cannot be written as a fraction of two integers.
Assume the opposite: that $\sqrt{2}$ *is* rational. Then we can write

$$\sqrt{2} = \frac{p}{q},$$

where p and q are integers **sharing no common factor**, so the fraction is fully cancelled. The contradiction at the end of the proof depends on this condition, so state it explicitly. Squaring both sides gives $2 = \frac{p^2}{q^2}$, hence

$$p^2 = 2q^2.$$

So p^2 is even, which forces p itself to be even (the square of an odd number is odd). Write $p = 2m$. Then $p^2 = 4m^2$, and substituting,

$$4m^2 = 2q^2 \implies q^2 = 2m^2,$$

so q^2 is even, and therefore q is even too. But now p and q are both even, sharing the factor 2, contradicting our choice of a fully cancelled fraction. The assumption that $\sqrt{2}$ is rational is impossible, so $\sqrt{2}$ is irrational. ■

EXAMPLE 3.10

Prove that there are infinitely many prime numbers. This argument is due to Euclid, around 300 BCE.
Assume the opposite: that there are only finitely many primes, and list them all as $p_1, p_2, \dots, p_n$. Build the number

$$N = p_1 \times p_2 \times \cdots \times p_n + 1,$$

the product of every prime, plus one. Dividing N by any prime on the list leaves remainder 1, so N is divisible by none of them. Yet every integer greater than 1 has at least one prime factor. That factor cannot be on our list, so our list was not complete, a contradiction. No finite list can hold all the primes, so there are infinitely many. ■

CAUTION *State your assumption clearly at the start ("assume …is rational") and name the contradiction when you reach it. A proof by contradiction with no clearly identified contradiction proves nothing.*

YOUR TURN 3.3 Proof by Contradiction HL

7. Prove each statement by contradiction.

(a) There is no largest integer. [2]
(b) There is no smallest positive real number. [3]

8. Let n be an integer. Prove each statement by contradiction.

(a) If n^2 is odd then n is odd. [3]

(b) If n^2 is even then n is even. [3]

9. Prove each statement by contradiction.

(a) There are no integers x and y with $2x + 4y = 5$. [2]

(b) The sum of a rational number and an irrational number is irrational. [3]

3.4 Proof by Induction HL

Picture an endless line of dominoes. You cannot knock over infinitely many by hand. But if you can show two things, that the first domino falls, and that *whenever* a domino falls it topples the next, then you know with certainty that every domino in the line will fall, however long the line.

Mathematical induction is exactly this idea, used to prove a statement P_n for every integer n from some starting value onward. We write the statement as P_n, with a subscript, because it is the n-th claim in an infinite list of claims $P_1, P_2, P_3, \ldots$, not a function being evaluated. You verify the first claim, then prove that each claim forces the one after it.

KEY **Principle of mathematical induction.** To prove that P_n holds for all integers $n \geq 1$:

1. Show that P_1 is true.
2. Assume that P_k is true for **some** $k \geq 1$ (the *inductive hypothesis*).
3. Show that P_k implies P_{k+1}.
4. Conclude: *since P_1 is true, and P_k true implies P_{k+1} true, by the principle of mathematical induction P_n is true for all integers $n \geq 1$.*

(If the statement starts later, at n_0, begin at P_{n_0} instead of P_1.)

Every induction proof follows these four steps, and the sentence in step 4 should be written out in full. It is not decoration. It is the step that turns the two facts you proved into a conclusion about every n, and it carries its own mark in examinations.

CAUTION *In step 2, assume P_k for **some** k, never for all k. Assuming it for all k assumes the very thing you are proving, and examiners withhold the reasoning marks for it. The hypothesis is one domino falling, not the whole line.*

Why induction works

The domino image can be made precise. Suppose steps 1 and 3 are done, and pick any n you like, say $n = 1000$. Then P_1 is true; P_1 forces P_2; P_2 forces P_3; and after 999 applications of the inductive step, P_{1000} is true. The same finite chain reaches *any* n, so no case escapes, even though you only ever proved two things.

There is a second way to see it, and it connects induction to the previous section. Suppose the two steps are done, yet P_n somehow failed for some values of n. Among the failures there would be a *smallest* one, say P_m. Now $m \neq 1$, because step 1 checked P_1. So $m-1 \geq 1$, and P_{m-1} is true, since m was the smallest failure. But step 3 says that P_{m-1} forces P_m, which contradicts the failure of P_m. So induction is itself a proof by contradiction, and the two techniques of this chapter are closely connected.

Induction is not only for sums. In this course it also proves divisibility statements and inequalities below, De Moivre's theorem for complex numbers in Ch. 18, and formulas for n-th derivatives in Ch. 14. Whenever a claim has the form "for every integer n", induction is the first method to consider.

Induction for a sum

How to approach it. Here P_n says that a sum equals a formula. Add the next term of the sum to both sides of the assumption P_k. Then rearrange the right-hand side until it becomes the original formula with $k+1$ written wherever n appeared. Write that target expression down before you begin, so you know what you are aiming at.

EXAMPLE 3.11

Prove that $1+2+3+\cdots+n = \dfrac{n(n+1)}{2}$ for all integers $n \geq 1$.

Let P_n be the statement $1+2+\cdots+n = \dfrac{n(n+1)}{2}$.

Step 1 (P_1): the left side is 1; the right side is $\dfrac{1\cdot 2}{2} = 1$. They agree, so P_1 is true.

Step 2: assume P_k is true for some $k \geq 1$, that is,

$$1+2+\cdots+k = \frac{k(k+1)}{2}.$$

Step 3: add the next term, $k+1$, to both sides:

$$1+2+\cdots+k+(k+1) = \frac{k(k+1)}{2} + (k+1).$$

Factor $(k+1)$ out of the right side:

$$= (k+1)\left(\frac{k}{2}+1\right) = (k+1)\cdot\frac{k+2}{2} = \frac{(k+1)(k+2)}{2}.$$

This is exactly P_{k+1}, the original formula with n replaced by $k+1$. So $P_k \Rightarrow P_{k+1}$.

Step 4: since P_1 is true, and P_k true implies P_{k+1} true, by the principle of mathematical induction P_n is true for all integers $n \geq 1$. ■

Induction for divisibility

The same four steps prove divisibility statements.

How to approach it. Here P_n says that an expression E_n is divisible by a fixed number d. Turn the assumption into an equation: divisible by d means $E_k = dm$ for some integer m. Then rewrite E_{k+1} so that E_k appears inside it, replace that part by dm, and take d outside as a common factor. The aim is to reach the form $d \times$ (integer).

EXAMPLE 3.12

Prove that $8^n - 1$ is divisible by 7 for all integers $n \geq 1$.

Let P_n be the statement "$8^n - 1$ is divisible by 7."

Step 1 (P_1): $8^1 - 1 = 7$, which is divisible by 7. So P_1 is true.

Step 2: assume P_k is true for some $k \geq 1$, so $8^k - 1 = 7m$ for some integer m.

Step 3: consider the next case:

$$8^{k+1} - 1 = 8 \cdot 8^k - 1 = 8(8^k - 1) + 8 - 1 = 8(7m) + 7 = 7(8m + 1).$$

Since $8m + 1$ is an integer, $8^{k+1} - 1$ is divisible by 7, which is P_{k+1}. So $P_k \Rightarrow P_{k+1}$.

Step 4: since P_1 is true, and P_k true implies P_{k+1} true, by the principle of mathematical induction P_n is true for all integers $n \geq 1$. ■

Notice where the hypothesis was used: rewriting $8^k - 1$ as $7m$ is the only reason the final line factors cleanly.

Induction for an inequality

How to approach it. Here P_n says that $A_n > B_n$. Inequalities are proved by chaining: if $a > b$ and $b \geq c$, then $a > c$. Start from A_{k+1} and rewrite it so that A_k appears, then use the assumption to replace A_k by something smaller. That gives the first link. Now prove separately that this new quantity is at least B_{k+1}, which gives the second link. The two links join into a single chain from A_{k+1} down to B_{k+1}.

EXAMPLE 3.13

Prove that $2^n > n^2$ for all integers $n \geq 5$. The base case is not $n = 1$ here, and that matters: the inequality is false for $n = 2, 3, 4$.

Let P_n be the statement $2^n > n^2$.

Step 1 (P_5): $2^5 = 32$ and $5^2 = 25$, and $32 > 25$. So P_5 is true.

Step 2: assume P_k is true for some $k \geq 5$, that is, $2^k > k^2$.

Step 3: then

$$2^{k+1} = 2 \cdot 2^k > 2k^2.$$

It is now enough to show $2k^2 \geq (k+1)^2$. Expanding, this asks whether $2k^2 \geq k^2+2k+1$, that is, $k^2-2k-1 \geq 0$. For $k \geq 5$ this certainly holds, since $k^2-2k-1 = (k-1)^2-2 \geq 16-2 > 0$.

Therefore

$$2^{k+1} > 2k^2 \geq (k+1)^2,$$

which is P_{k+1}. So $P_k \Rightarrow P_{k+1}$.
Step 4: since P_5 is true, and P_k true implies P_{k+1} true for $k \geq 5$, by the principle of mathematical induction $2^n > n^2$ for all integers $n \geq 5$. ■

CAUTION *Two ways an induction proof fails: forgetting the base case (the dominoes can topple each other yet never start), or proving the inductive step without using the inductive hypothesis. If you never used the assumption P_k, you have not built a chain, only restated the problem.*

YOUR TURN 3.4 Proof by Induction HL

10. Prove each result by induction, for all $n \geq 1$.

(a) $1 + 2 + 3 + \cdots + n = \dfrac{n(n+1)}{2}$ [4]
(b) $1 + 3 + 5 + \cdots + (2n-1) = n^2$ [4]

11. Prove each result by induction, for all $n \geq 1$.

(a) $2 + 4 + 6 + \cdots + 2n = n(n+1)$ [4]
(b) $1 + 2 + 4 + \cdots + 2^{n-1} = 2^n - 1$ [4]

12. Prove each divisibility result by induction, for all $n \geq 1$.

(a) $3^n - 1$ is divisible by 2. [3]
(b) $4^n - 1$ is divisible by 3. [3]
(c) $5^n - 1$ is divisible by 4. [3]

13.

(a) Prove by induction that $2^n \geq n + 1$ for all $n \geq 0$. [3]
(b) Explain why the base case here is $n = 0$ rather than $n = 1$. [2]

Chapter 3 · Proofs

Reflections

TOK A proof reduces a claim to things already accepted, but those starting points, the axioms, are themselves assumed, not proved. In 1931 Kurt Gödel showed that any logical system rich enough to describe ordinary arithmetic must contain true statements it cannot prove. Mathematical certainty, then, is real but not absolute: it is certainty *relative* to a chosen foundation. Does this make mathematics more like a discovered landscape or a built structure?

IM Induction was not invented in one place or time. A recognisable form appears in the work of the Persian mathematician al-Karaji around 1000 CE, who used it to establish results about sums of powers. The method was developed further by Levi ben Gerson in the fourteenth century and by Pascal in the seventeenth, and the name "mathematical induction" was finally fixed by Augustus De Morgan in 1838. The proof that primes are infinite is older still, written down by Euclid in Alexandria around 300 BCE, and it is still taught in his original form more than two thousand years later.

RW Proof is not confined to pure mathematics. In software engineering, *formal verification* proves that a program meets its specification exactly, a guarantee no amount of testing can give, used where failure is catastrophic: aircraft control, medical devices, spacecraft. Modern cryptography depends on proved properties of prime numbers. The security of online banking is, in the end, a theorem.

EXT In 1976 the four-colour theorem, that any map can be coloured with four colours so no two neighbouring regions match, became the first major theorem whose proof required a computer. The machine checked nearly two thousand separate cases, far more than a person could check by hand. This raised a new question: if a proof is too long for any human to read, but every step has been verified by a machine, is it still a proof? What exactly is a proof *for*, certainty, or understanding?

The language of proof

Mathematics has a precise vocabulary for talking about statements and their truth. The terms below appear throughout this book and in any mathematical reading; the symbolic forms ($\neg$, $\Rightarrow$, $\iff$, contrapositive) are language for reading and for ToK discussion rather than examinable AA content.

Term	Meaning	Example
statement	a sentence that is definitely true or false	"7 is prime"
conjecture	believed true, not yet proved	every even $n > 2$ is a sum of two primes
theorem	a statement that has been proved	$\sqrt{2}$ is irrational
axiom	accepted without proof as a starting point	$a + b = b + a$ for real numbers
lemma	a small result proved on the way to a bigger one	"if n^2 is even then n is even," inside the $\sqrt{2}$ proof
corollary	an easy consequence of a theorem	$3 + \sqrt{2}$ is irrational
negation $\neg P$	the opposite statement	$\neg(n$ is even$)$: "n is odd"
implication $P \Rightarrow Q$	if P holds, Q must hold	n divisible by 4 $\Rightarrow$ n even
converse $Q \Rightarrow P$	the reversed implication, **not** automatic	n even $\Rightarrow$ divisible by 4 is *false* $(n = 6)$
contrapositive $\neg Q \Rightarrow \neg P$	logically identical to $P \Rightarrow Q$	n odd $\Rightarrow$ n not divisible by 4
equivalence $P \iff Q$	each implies the other	n even $\iff$ n^2 even
counterexample	one case that disproves a universal claim	$n = 40$ for "$n^2 + n + 41$ is prime"
identity $\equiv$	equal for every value of the variable	$(x + 1)^2 \equiv x^2 + 2x + 1$
■ (or QED)	marks the end of a proof	—

Note that the converse row contains its own counterexample. Reversing a true implication produces a new claim, which needs its own proof, and $n = 6$ shows that this particular one is false.

A gallery of conjectures

Every theorem started as a conjecture. Here are eight famous ones, each simple enough to state in a sentence, together with what has happened to them so far. None of them is examinable. Together they show what this chapter is about, in real mathematics.

Goldbach conjecture (1742) — open. Every even number greater than 2 is the sum of two primes: $28 = 5 + 23$, $100 = 3 + 97$. Verified for every even number up to 4×10^{18}, and still unproven. The most famous "obviously true, nobody can prove it" statement in mathematics.

Twin prime conjecture — open. There are infinitely many primes p for which $p + 2$ is also prime, like $11, 13$ and $101, 103$. In 2013 Yitang Zhang proved that there are infinitely many prime pairs at most 70 million apart. Later work reduced that distance to 246. This is much closer to the conjecture, but 246 is still not 2.

Collatz conjecture — open. Take any positive integer; halve it if even, triple it and add 1 if odd; repeat. Does every starting number eventually reach 1? Try 27 on your GDC: it takes 111 steps and climbs to 9232 on the way. Verified for astronomically many starting values, proved for none of them collectively. Erdős said mathematics is "not yet ripe" for it.

Odd perfect numbers — open for ∼2300 years. A number is perfect if it equals the sum of its proper divisors, like $6 = 1 + 2 + 3$ and $28 = 1 + 2 + 4 + 7 + 14$. Every perfect number ever found is even. Whether an odd one exists is arguably the oldest unsolved problem in mathematics; any example must exceed 10^{1500}.

Legendre's conjecture — open. Between any two consecutive squares n^2 and $(n + 1)^2$ there is always at least one prime. True in every case ever checked, and still unproved.

Fermat's Last Theorem (1637) — proved, 1995. No positive integers satisfy $x^n + y^n = z^n$ for $n \geq 3$. Fermat wrote in the margin of a book that he had a proof, but that the margin was too small to hold it. The first accepted proof came 358 years later, from Andrew Wiles, who worked on it for seven years in near secrecy, and it uses mathematics far beyond anything available to Fermat. Conjectures can be settled, and this one was.

Pólya's conjecture (1919) — disproved. At least half of the numbers up to any N have an odd number of prime factors. Verified for millions of cases and widely believed for forty years, until it was shown to fail, with smallest known counterexample near $N = 906{,}150{,}257$. This is the circle problem from the start of the chapter, on a far larger scale: millions of cases agreed with the claim, one case did not, and the claim is false.

TOK Which of these deserves more confidence: Goldbach, verified for 4×10^{18} cases but unproven, or a theorem proved once and checked by a handful of humans? Pólya's conjecture had millions of confirmations and was false. What, then, is evidence actually worth in mathematics, and is that different from every other field of knowledge?

End-of-Chapter Problems — Chapter 3: Proofs

1. Prove that the product of any three consecutive integers is divisible by 6. [5]

2. Consider the cube of a binomial.

(a) Prove the identity $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$. [3]

(b) Hence prove that $(n+1)^3 - n^3 = 3n^2 + 3n + 1$ for every integer n. [3]

3. Disprove the claim "$n^2 + 1$ is never divisible by 5" by giving two counterexamples. [3]

4. Prove by contradiction that $\sqrt{3}$ is irrational. [6]

5. Let a be a non-zero rational number and b an irrational number.

(a) Prove by contradiction that ab is irrational. [4]

(b) Hence state, with a reason, whether $\dfrac{\sqrt{2}}{5}$ is rational or irrational. [2]

6.

(a) Prove by contradiction that $x^2 - 6x + 10 = 0$ has no real solution, without using the discriminant. [3]

(b) Prove by contradiction that if n is an integer and n^2 is a multiple of 3, then n is a multiple of 3. [3]

7. Prove that $\sqrt{2} + \sqrt{3}$ is irrational. You may assume that $\sqrt{6}$ is irrational. [6]

8. Prove by contradiction that $\sqrt[3]{5}$ is irrational. [6]

9. Prove by induction that $7^n - 1$ is divisible by 6 for all $n \geq 1$. [5]

10. Prove by induction that $6^n + 4$ is divisible by 5 for all $n \geq 1$. [5]

11. Prove by induction that $n! > 2^n$ for all $n \geq 4$, stating why the base case is $n = 4$. [5]

12. Prove by induction that $5^{2n} - 1$ is divisible by 24 for all $n \geq 1$. [5]

13. Prove by induction that $2^n > n$ for all $n \geq 1$. [5]

14. Prove by induction that $9^n - 1$ is divisible by 8 for all $n \geq 1$. [5]

15. Prove by induction that $n^3 + 2n$ is divisible by 3 for all $n \geq 1$. [5]

16. Prove by induction that $\sum_{r=1}^{n} r^2 = \dfrac{n(n+1)(2n+1)}{6}$ for all $n \geq 1$. [6]

17. Prove by induction that $\sum_{r=1}^{n} r^3 = \dfrac{n^2(n+1)^2}{4}$ for all $n \geq 1$. [6]

18. Prove by induction that $\sum_{r=1}^{n} r(r+1) = \dfrac{n(n+1)(n+2)}{3}$ for all $n \geq 1$. [6]

19. Prove by induction that $\sum_{r=1}^{n} (3r-2) = \dfrac{n(3n-1)}{2}$ for all $n \geq 1$. [6]

20. Prove by induction that $\sum_{r=1}^{n} \dfrac{1}{r(r+1)} = \dfrac{n}{n+1}$ for all $n \geq 1$. [6]

21. A student claims that $n^2 - n + 11$ is prime for every positive integer n.

(a) Verify the claim for $n = 1$ and $n = 2$. [2]
(b) Disprove the claim with a counterexample. [3]

22. Consider parity of squares.

(a) Prove that if n is odd then n^2 is odd. [2]
(b) Hence prove by contradiction that if n^2 is even then n is even. [3]

23. Prove that $1 + 3 + 5 + \cdots + (2n-1) = n^2$ in two ways:

(a) directly, using the sum of an arithmetic sequence; [3]
(b) by induction. [3]

24. Prove that $\log_2 3$ is irrational. [6]

25. Prove by induction that $\sum_{r=1}^{n} r \cdot r! = (n+1)! - 1$ for all $n \geq 1$. [6]

26.

(a) Prove by induction that $\sum_{r=1}^{n} r = \dfrac{n(n+1)}{2}$ for all $n \geq 1$. [5]
(b) You may assume that $\sum_{r=1}^{n} r^2 = \dfrac{n(n+1)(2n+1)}{6}$. Find $\sum_{r=1}^{n} r(r-1)$, giving your answer in fully factorised form. [4]
(c) Hence evaluate $\sum_{r=1}^{20} r(r-1)$. [3]

(d) Show that $\sum_{r=1}^{n} r(r-1) = 2\binom{n+1}{3}$, and explain why this means the sum is always even. [3]

27.

(a) Prove that if n^2 is even, then n is even. [3]

(b) Hence prove by contradiction that $\sqrt{2}$ is irrational. [5]

(c) Explain precisely where the argument in part (b) breaks down if you attempt to use it to prove that $\sqrt{4}$ is irrational. [3]

(d) A student writes: *"$\sqrt{2}$ is irrational, so $\sqrt{2}\times\sqrt{2}$ must be irrational too."* Explain the error, and state what it shows about products of irrational numbers. [3]

28. A sequence is defined by $u_1 = 1$ and $u_{n+1} = \dfrac{u_n}{1+u_n}$ for $n \geq 1$.

(a) Find u_2, u_3 and u_4. [3]

(b) Write down a conjecture for u_n in terms of n. [1]

(c) Prove your conjecture by induction. [5]

(d) A student's whole proof reads: *"Assume the statement is true for $n = k$. Then it is true for $n = k+1$. Hence it is true for all n."* State what is missing, and explain why the argument establishes nothing without it. [3]

(e) Give a statement $P(n)$ for which $P(k) \Rightarrow P(k+1)$ is true for every k, but $P(n)$ is false for every n. Justify both claims. [3]

Challenge Questions

29. **Counting the regions of a circle.** This investigation completes the problem from the start of the chapter. Throughout, n points are placed on a circle so that no three chords meet at a single interior point.

(a) By drawing, verify that $n = 1, 2, 3, 4, 5$ give $1, 2, 4, 8, 16$ regions. [2]

(b) Explain why each interior intersection point corresponds to exactly one choice of 4 of the n points, and deduce that there are $\binom{n}{4}$ interior points. [3]

(c) The chords and arcs form a network. Show that it has $V = n + \binom{n}{4}$ vertices and $E = n + \binom{n}{2} + 2\binom{n}{4}$ edges. *(Hint: each interior point adds one extra segment to each of the two chords through it)* [4]

(d) Euler's formula for a connected planar network states $V - E + F = 2$, where F counts all faces including the region outside the circle. Use it to prove that the number of regions inside the circle is $\binom{n}{4} + \binom{n}{2} + 1$. [4]

(e) Hence find the number of regions for $n = 6$ and $n = 10$. [2]

(f) If the n points are equally spaced, the count for $n = 6$ is 30 rather than 31. Explain what changes. [3]

30. Fermat's conjecture. In 1640 Fermat studied $F_n = 2^{2^n} + 1$.

(a) Show that F_0, F_1, F_2, F_3 and F_4 are prime. [3]

(b) Suppose m has an odd factor $d > 1$, so $m = de$. Using the factorisation

$$x^d + 1 = (x+1)\left(x^{d-1} - x^{d-2} + \cdots - x + 1\right) \quad (d \text{ odd}),$$

with $x = 2^e$, prove that $2^m + 1$ is not prime. [5]

(c) Deduce that if $2^m + 1$ is prime, then m must be a power of 2. [2]

(d) Verify that 641 divides $F_5 = 4\,294\,967\,297$, and state what this shows about Fermat's conjecture. [3]

31. A proof that names no example. Consider the number $\sqrt{2}^{\sqrt{2}}$.

(a) Suppose $\sqrt{2}^{\sqrt{2}}$ is rational. Explain why this immediately produces an irrational number raised to an irrational power that is rational. [2]

(b) Suppose instead $\sqrt{2}^{\sqrt{2}}$ is irrational. Show that $\left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = 2$, and explain why this also produces such a pair. [4]

(c) Conclude that there exist irrational a and b with a^b rational. [2]

(d) This argument proves that such a pair exists without telling you which case is the true one. Compare it with the proof that $\sqrt{2}$ is irrational, and comment on what each proof gives you. [4]

32. Fibonacci identities. The Fibonacci numbers are defined by $F_1 = F_2 = 1$ and $F_{n+2} = F_{n+1} + F_n$.

(a) Prove by induction that $F_1 + F_2 + \cdots + F_n = F_{n+2} - 1$ for all $n \geq 1$. [5]

(b) Prove by induction that $F_1^2 + F_2^2 + \cdots + F_n^2 = F_n F_{n+1}$ for all $n \geq 1$. [5]

(c) Part (b) has a picture proof: squares of sides $F_1, F_2, \ldots, F_n$ tile a rectangle. State the dimensions of that rectangle and explain how it proves the identity. [3]

33. Divisibility by consecutive integers.

(a) Prove that $n^3 - n$ is divisible by 6 for every integer n. [3]

(b) Prove that $n^5 - n$ is divisible by 30 for every integer n. *(Hint: factor $n^5 - n$ and consider divisibility by 2, 3 and 5 separately)* [6]

(c) Investigate whether $n^7 - n$ is divisible by 42 for every integer n. State the general pattern you notice about $n^p - n$ when p is prime. [5]